SIX EQUIVALENT FORMULATIONS OF RELATIVE AMPLENESS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-morphisms/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot. Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Editorial note. Screening note: the proof contains 'It is immediate from the definition that (1) implies (2) and (2) implies (3)', which is one legitimate implication inside a long argument; the author proves the remaining implications in full, so this is not a hint-only proof.

Allowed standard prerequisites (editorial).

- Ample invertible sheaves on a scheme over an affine base are separated, and the relative Proj construction. Properties, Lemma \ref{properties-lemma-ample-separated}, Constructions, Section \ref{constructions-section-invertible-relative-proj}, and Constructions, Lemma \ref{constructions-lemma-invertible-map-into-relative-proj}, cited at proof lines 8719, 8735-8738, and 8743-8746. Provide the constructions used throughout.
- Ampleness descends and ascends along the constructions used, and ample sheaves restrict to closed subschemes. Properties, Lemma \ref{properties-lemma-open-in-proj-ample}, Properties, Lemma \ref{properties-lemma-ample-on-closed}, and Properties, Lemma \ref{properties-lemma-ample-power-ample}, cited at proof lines 8788-8795. Close the cycle at the end of the proof.

Source reference. Original lemma 01VJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2448–2448. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01O9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2186–2186. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 09MP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12187–12187. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01LC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2081–2081. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01O8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2185–2185. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01PZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2248–2248. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01Q1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2250–2250. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01O0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2177–2177. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01NK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2161–2161. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01Q2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2251–2251. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01PT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2242–2242. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01MW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2137–2137. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01PU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2243–2243. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original reference cite-EGA: sources/stacks-morphisms/pinned-original-source.tex, source lines 8679–8679. Bibliographic pointer appearing in the counted text; it carries no mathematical argument.

Lemma 0.2. *Let $f : X \to S$ be a quasi-compact morphism of schemes. Let $\mathcal{L}$ be an invertible sheaf on X. The following are equivalent:*

1. *The invertible sheaf $\mathcal{L}$ is f-ample.*
2. *There exists an open covering $S = \bigcup V_i$ such that each $\mathcal{L}|_{f^{-1}(V_i)}$ is ample relative to $f^{-1}(V_i) \to V_i$.*
3. *There exists an affine open covering $S = \bigcup V_i$ such that each $\mathcal{L}|_{f^{-1}(V_i)}$ is ample.*
4. *There exists a quasi-coherent graded $\mathcal{O}_S$-algebra $\mathcal{A}$ and a map of graded $\mathcal{O}_X$-algebras $\psi : f^*\mathcal{A} \to \bigoplus_{d\geq 0} \mathcal{L}^{\otimes d}$ such that $U(\psi) = X$ and*
$$r_{\mathcal{L},\psi} : X \longrightarrow \underline{Proj}_S(\mathcal{A})$$
is an open immersion (see Constructions, Lemma 01O9 for notation).
5. *The morphism f is quasi-separated and part (4) above holds with $\mathcal{A} = f_*(\bigoplus_{d\geq 0} \mathcal{L}^{\otimes d})$ and ψ the adjunction mapping.*
6. *Same as (4) but just requiring $r_{\mathcal{L},\psi}$ to be an immersion.*

Proof. It is immediate from the definition that (1) implies (2) and (2) implies (3). It is clear that (5) implies (4).

Assume (3) holds for the affine open covering $S = \bigcup V_i$. We are going to show (5) holds. Since each $f^{-1}(V_i)$ has an ample invertible sheaf we see that $f^{-1}(V_i)$ is separated (Properties, Lemma 09MP). Hence f is separated. By Schemes, Lemma 01LC we see that $\mathcal{A} = f_*(\bigoplus_{d\geq 0} \mathcal{L}^{\otimes d})$ is a quasi-coherent graded $\mathcal{O}_S$-algebra. Denote $\psi : f^*\mathcal{A} \to \bigoplus_{d\geq 0} \mathcal{L}^{\otimes d}$ the adjunction mapping. The description of the open $U(\psi)$ in Constructions, Section 01O8 and the definition of ampleness of $\mathcal{L}|_{f^{-1}(V_i)}$ show that $U(\psi) = X$. Moreover, Constructions, Lemma 01O9 part (3) shows that the restriction of $r_{\mathcal{L},\psi}$ to $f^{-1}(V_i)$ is the same as the morphism from Properties, Lemma 01PZ which is an open immersion according to Properties, Lemma 01Q1. Hence (5) holds.

Let us show that (4) implies (1). Assume (4). Denote $\pi : \underline{\mathrm{Proj}}_S(\mathcal{A}) \to S$ the structure morphism. Choose $V \subset S$ affine open. By Constructions, Definition 01O0 we see that $\pi^{-1}(V) \subset \underline{\mathrm{Proj}}_S(\mathcal{A})$ is equal to $\mathrm{Proj}(A)$ where $A = \mathcal{A}(V)$ as a graded ring. Hence $r_{\mathcal{L},\psi}$ maps $f^{-1}(V)$ isomorphically onto a quasi-compact open of $\mathrm{Proj}(A)$. Moreover, $\mathcal{L}^{\otimes d}$ is isomorphic to the pullback of $\mathcal{O}_{\mathrm{Proj}(A)}(d)$ for some $d \geq 1$. (See part (3) of Constructions, Lemma 01O9 and the final statement of Constructions, Lemma 01NK.) This implies that $\mathcal{L}|_{f^{-1}(V)}$ is ample by Properties, Lemmas 01Q2 and 01PT.

Assume (6). By the equivalence of (1) - (5) above we see that the property of being relatively ample on X/S is local on S. Hence we may assume that S is affine, and we have to show that $\mathcal{L}$ is ample on X. In this case the morphism $r_{\mathcal{L},\psi}$ is identified with the morphism, also denoted $r_{\mathcal{L},\psi} : X \to \mathrm{Proj}(A)$ associated to the map $\psi : A = \mathcal{A}(V) \to \Gamma_*(X, \mathcal{L})$. (See references above.) As above we also see that $\mathcal{L}^{\otimes d}$ is the pullback of the sheaf $\mathcal{O}_{\mathrm{Proj}(A)}(d)$ for some $d \geq 1$. Moreover, since X is quasi-compact we see that X gets identified with a closed subscheme of a quasi-compact open subscheme $Y \subset \mathrm{Proj}(A)$. By Constructions, Lemma 01MW (see also Properties, Lemma 01Q2) we see that $\mathcal{O}_Y(d')$ is an ample invertible sheaf on Y for some $d' \geq 1$. Since the restriction of an ample sheaf to a closed subscheme is ample, see Properties, Lemma 01PU we conclude that the pullback of $\mathcal{O}_Y(d')$ is

ample. Combining these results with Properties, Lemma 01PT we conclude that $\mathcal{L}$ is ample as desired. $\square$